

SocialPilot – Social Media Scheduler & Campaign Management Platform

Testing & Deployment Report

Project: SocialPilot – Social Media Scheduler & Campaign Management Platform

Role: Testing & Deployment

1. Project Overview

The SocialPilot – Social Media Scheduler & Campaign Management Platform was developed to provide a centralized solution for scheduling, managing, publishing, and monitoring content across multiple social media platforms. The platform allows users to create campaigns, schedule posts, monitor publishing activities, and analyze campaign performance through a single dashboard. The project was designed to improve productivity by reducing manual posting efforts and providing automated scheduling with real-time analytics. Throughout the development process, emphasis was placed on ensuring system reliability, scalability, and seamless deployment.

As the Testing & Deployment team member, my responsibility was to verify that every module functioned correctly before deployment. I conducted functional testing, API validation, integration testing, performance verification, deployment validation, and final production testing to ensure that the application was stable and ready for end users.

2. Objectives of Testing & Deployment

The primary objective of my role was to ensure that the SocialPilot platform was reliable, secure, and deployment-ready. My work included validating all project modules, verifying backend APIs, checking system integration, identifying defects, and confirming that all functionalities worked correctly in the deployment environment.

The objectives achieved included:

- Tested all implemented project modules.
- Validated backend API responses using Swagger.
- Verified authentication and authorization.
- Performed integration testing between frontend and backend.

- Executed deployment using Docker.
- Verified production configuration.
- Conducted final workflow validation.
- Prepared deployment documentation.
- Confirmed successful project completion.

3. Milestone 1 – Testing Environment Preparation

During the first milestone, I prepared the complete testing environment after the project setup was completed. The backend server, database connections, authentication modules, and project dependencies were verified before testing activities began.

Activities Completed

- Verified project installation.
- Configured testing environment.
- Tested database connectivity.
- Validated user authentication.
- Verified role-based access control.
- Tested account management workflows.
- Checked API documentation using Swagger.
- Verified project initialization.

Outcome

The complete project environment was successfully validated, and all initial modules were confirmed to be functioning correctly before further development.

4. Milestone 2 – Functional Testing & API Validation

After the scheduling and publishing modules were completed, extensive functional testing was performed.

Activities Completed

- Tested content scheduling.
- Verified publishing calendar.
- Tested recurring scheduling.
- Validated draft management.
- Tested publishing queue.

- Verified publishing logs.
- Checked automated scheduling.
- Performed API testing using Swagger.
- Validated response codes.
- Tested invalid input handling.
- Verified error messages.

Testing Performed

Authentication Testing

- Login validation
- Registration testing
- Invalid credential testing

Scheduling Testing

- Future scheduling
- Date validation
- Queue verification

Publishing Testing

- Scheduled publishing
- Failed publishing
- Retry mechanism

Outcome

All APIs responded correctly, scheduling worked successfully, publishing queues were validated, and system functionality was confirmed.

5. Milestone 3 – Integration Testing & Performance Validation

After campaign management and analytics modules were completed, integration testing was performed across all project modules.

Activities Completed

- Tested campaign workflows.
- Verified analytics dashboard.
- Validated reporting module.

- Tested audience analytics.
- Checked engagement tracking.
- Verified notification system.
- Tested report generation.
- Performed integration testing.
- Verified database consistency.
- Tested API integration.

Performance Testing

The following parameters were evaluated:

- API response time
- Database query execution
- Dashboard loading speed
- Concurrent user handling
- Background task execution
- Scheduling accuracy

Outcome

All integrated modules worked successfully without major issues, and the application demonstrated stable performance.

6. Milestone 4 – Deployment & Final Validation

The final stage focused on deployment and production validation.

Activities Completed

- Configured Docker environment.
- Built Docker containers.
- Tested Docker Compose.
- Verified environment variables.
- Configured production settings.
- Started backend services.
- Tested deployed APIs.
- Validated deployment logs.
- Performed smoke testing.
- Conducted regression testing.

- Completed final documentation.

Deployment Process

1. Built Docker images.
2. Configured application environment.
3. Started backend services.
4. Connected database.
5. Verified API endpoints.
6. Performed deployment testing.
7. Checked production logs.
8. Validated complete workflow.

Outcome

The SocialPilot platform was successfully deployed and validated. All core functionalities worked correctly in the deployment environment.

```
D:\SocialPilot\backend>uvicorn main:app --reload
INFO:      Will watch for changes in these directories: ['D:\SocialPilot\backend']
INFO:      Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:      Started reloader process [15632] using StatReload
INFO:      Started server process [18628]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
```

Figure 1: Backend Server Started Successfully

```
PS D:\SocialPilot\backend> curl -X POST http://127.0.0.1:8000/api/v1/auth/login ^
>> -H "Content-Type: application/json" ^
>> -d {"email\":\"testuser@gmail.com\",\"password\":\"test123\"}
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9....",
  "token_type": "bearer",
  "user": {
    "id": 2,
    "full_name": "Test User",
    "email": "testuser@gmail.com",
    "role": "user"
  }
}
PS D:\SocialPilot\backend>
```

Figure 2: API Login Test Executed Successfully

7. Testing Methodology

The following testing methods were completed during the project:

Functional Testing

Verified that every project module performed according to requirements.

API Testing

Validated backend endpoints using Swagger and confirmed successful responses.

Integration Testing

Verified communication between frontend, backend, database, and external services.

Performance Testing

Measured system performance under normal operating conditions.

Regression Testing

Ensured that newly implemented features did not affect existing functionality.

Deployment Testing

Validated complete application functionality after deployment.

8. Tools Used

The following tools were used during Testing & Deployment:

- Python
- FastAPI
- Swagger/OpenAPI
- SQLAlchemy
- PostgreSQL
- Docker
- Docker Compose
- Git & GitHub
- Visual Studio Code
- Postman
- Redis

- Uvicorn

These tools supported testing, deployment, API validation, monitoring, and production verification.

9. Challenges Faced & Solutions

Challenge	Solution
API validation issues	Verified request formats and corrected endpoint configuration.
Deployment configuration errors	Updated Docker configuration and environment variables.
Database connection issues	Verified connection strings and migration status.
Authentication failures	Tested JWT configuration and corrected token validation.
Integration inconsistencies	Performed end-to-end workflow testing and resolved API communication issues.

10. Conclusion

The Testing & Deployment activities were successfully completed for the SocialPilot – Social Media Scheduler & Campaign Management Platform. All functional modules were thoroughly tested, APIs were validated, deployment procedures were completed, and the final production environment was verified successfully.

The project achieved stable system performance, successful deployment, reliable API functionality, secure authentication, and complete workflow validation. Through this role, valuable practical experience was gained in software testing, API validation, Docker deployment, production configuration, system verification, and professional software delivery. The project was successfully completed and confirmed to be ready for production use.